QIUZHEN QUALIFY EXAM FOR ARTIFICIAL INTELLIGENCE

MOCK EXAMINATION NO. 5

Instructions

- This examination consists of four sections, each worth 33 points. Choose exactly three sections and answer all questions in the selected sections.
- Write your name and student ID on the answer sheet for 1 additional point. The maximum possible score is 100.
- State the three selected sections at the beginning of the answer sheet. If more than three sections are attempted, only the first three will be graded.
- Unless stated otherwise, justify every claim and show the essential derivation. Unsupported final formulas receive limited credit.

A. Machine Learning Theory

1. [8 pts.] Let $\mathcal{H}_1, \mathcal{H}_2, \ldots$ be finite binary hypothesis classes with $|\mathcal{H}_j| = M_j$. Let positive numbers π_j satisfy $\sum_j \pi_j \le 1$. Under zero-one loss, derive a simultaneous high-probability bound for every $h \in \mathcal{H}_j$ and every j. Use it to define a structural-risk-minimization rule and prove an oracle inequality comparing its population risk with the best hypothesis in every class. State why a summable allocation π_j is needed.

2. [8 pts.] For K-class softmax regression, $W \in \mathbb{R}^{K\times d}$, $z = Wx$, $p = \text{softmax}(z)$, and $\ell(W; x, y) = -\log p_y$. Derive the gradient and Hessian with dimensions, prove convexity in W, and identify a nontrivial family of parameter transformations that leaves all class probabilities unchanged. Explain the consequences for strict convexity and give two mathematically valid ways to restore identifiability or uniqueness.

3. [8 pts.] Let $\mathcal{F} = \{x \mapsto w^\top x : \|w\|_2 \le B\}$ with $\|x\|_2 \le R$. For margin parameter $\gamma > 0$, define a bounded ramp loss that upper-bounds the binary zero-one loss and is $1/\gamma$-Lipschitz in the signed margin. Starting from a Rademacher-complexity bound, derive a high-probability population classification-error bound in terms of the empirical margin loss. Explain how the bound expresses a tradeoff in choosing γ.

4. [9 pts.] Let $A \in \mathbb{R}^{q\times d}$ satisfy the restricted isometry property of order $2s$ with constant $\delta_{2s} < 1$. Measurements obey $y = Ax + e$, where x is s-sparse and $\|e\|_2 \le \eta$. Suppose an estimator returns an s-sparse $\hat{x}$ satisfying $\|A\hat{x} - y\|_2 \le \eta$. Prove an explicit ℓ_2 error bound for $\hat{x}$. State what the result becomes when $\eta = 0$, and explain why this argument does not by itself analyze a nonsparse output of basis-pursuit denoising.

B. Deep Learning and Reinforcement Learning

1. [8 pts.] An image feature map $X \in \mathbb{R}^{H\times W\times C}$ is processed by a 3×3 convolution with C' output channels, stride 1 and padding 1. It is then flattened to $L = HW$ tokens, projected to a model width d, and passed through h-head self-attention with per-head key/value width $d_h = d/h$. Compute all principal tensor shapes, the convolution and attention parameter counts excluding biases, and leading-order time and activation-memory complexities. Analyze which parts are translation equivariant before and after adding absolute positional embeddings.

2. [9 pts.] Consider the forward diffusion SDE

$$dX_t = f(X_t, t)\,dt + g(t)\,dW_t$$

with smooth positive density p_t. Derive its Fokker–Planck equation. Then derive a deterministic probability-flow ODE involving the score $\nabla_x \log p_t(x)$ and prove that it has the same one-time marginal densities as the SDE. State the regularity assumptions and explain what fails when the learned score is inaccurate.

3. [8 pts.] Trajectories are sampled from a behavior policy μ, but the objective is the discounted return $J(\theta)$ of a target policy π_θ. Derive an off-policy policy-gradient estimator using per-decision importance ratios. State support and critic assumptions under which it is unbiased. Analyze the effect of clipping the ratios and explain why an off-policy actor–critic may still be unstable with nonlinear function approximation.

4. [8 pts.] Let two images be $I_1, I_2 : \Omega \to \mathbb{R}$, and let a dense optical-flow field $u(x) \in \mathbb{R}^2$ warp I_2 toward I_1. Formulate a differentiable photometric objective with a spatial regularizer, derive the gradient of the data term with respect to $u(x)$, and obtain the linearized brightness-constancy constraint. Explain the aperture problem and modify the objective to handle occlusion without permitting a trivial all-occluded solution.

C. Optimization Methods in Artificial Intelligence

1. [11 pts.] Let $f(x) = n^{-1} \sum_{i=1}^{n} f_i(x)$. At a fixed x, sample $I = i$ with probability $p_i > 0$ and define

$$g(x, I) = \frac{\nabla f_I(x)}{n p_I}.$$

(a) **[3 pts.]** Prove unbiasedness and derive the exact conditional variance.

(b) **[4 pts.]** Find the distribution p minimizing the conditional second moment, including zero-gradient cases.

(c) **[2 pts.]** For a minibatch of B independent samples with replacement, derive the variance of the averaged estimator.

(d) **[2 pts.]** Explain how usable upper bounds on component-gradient norms can define a fixed sampling scheme and identify the cost of estimating adaptive probabilities.

2. [10 pts.] Let $f(x) = \frac{1}{2} x^\top Q x$ with eigenvalues in $[\mu, L]$, $0 < \mu \le L$, and $\kappa = L/\mu$.

(a) **[4 pts.]** Derive the optimal constant-step worst-case contraction factor of gradient descent.

(b) **[4 pts.]** For the strongly-convex Nesterov iteration with step $1/L$ and momentum $\beta = (\sqrt{L} - \sqrt{\mu})/(\sqrt{L} + \sqrt{\mu})$, diagonalize the recurrence and determine its worst-case asymptotic factor, noting any repeated-root transient.

(c) **[2 pts.]** Compare their iteration complexities and state why the quadratic spectral calculation is not a proof for arbitrary nonconvex objectives.

3. [6 pts.] Let $D_t \succ 0$ be Adam's diagonal normalization matrix and α_t its learning rate. Compare the coupled L_2 update

$$w_{t+1} = w_t - \alpha_t D_t (\nabla f_t(w_t) + \lambda w_t)$$

with AdamW's decoupled weight decay

$$w_{t+1} = (1 - \alpha_t \lambda) w_t - \alpha_t D_t \nabla f_t(w_t).$$

Determine exactly when the two updates are equivalent after a scalar reparameterization of λ, and explain the coordinate-wise difference otherwise.

4. [6 pts.] Consider

$$\min_{w} \frac{1}{2} \|Aw - b\|_2^2 + \lambda \|w\|_1 \quad \text{subject to } Cw = d.$$

Derive the KKT conditions, including the complete coordinate-wise ℓ_1 subgradient. Give one explicit primal–dual proximal update and state why independently applying soft thresholding and Euclidean projection need not equal the proximal map of the constrained nonsmooth term.

D. Natural Language Processing

1. [7 pts.] A knowledge-aware embedding model uses the TransE score $s(h, r, t) = -\|e_h + e_r - e_t\|_2^2$ and also observes word–entity context pairs trained by SGNS. Formulate a joint negative-sampling objective for one positive triple and one positive word–entity pair. Derive the gradients of the triple term with respect to e_h, e_r, e_t. Explain type-aware negative sampling and one conflict that can arise between graph and text objectives.

2. [8 pts.] A long-context Transformer has L ordinary tokens, a symmetric local window of width w, and g global tokens that attend to all positions and are attended to by all positions. Derive the attention time and score-memory complexities. Determine the maximum number of layers needed for any ordinary token to influence any other token with and without global tokens. Discuss how causal masking changes the reachability statement.

3. [6 pts.] In a simplified model of in-context learning, a prompt contains examples $y_i = x_i^\top \theta + \varepsilon_i$, with prior $\theta \sim \mathcal{N}(0, \tau^2 I)$ and noise $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. Derive the posterior mean and covariance of θ and the posterior predictive distribution for a new x_*. Prove that the result is invariant to the ordering of demonstration pairs and state why an actual Transformer need not have this invariance.

4. [12 pts.] Design a citation-grounded retrieval-augmented generation system under both a context-token budget and a persistent-memory budget. Specify the retriever and generator objectives, hard-negative construction, evidence selection, citation alignment, and abstention. Model multi-turn retrieval/memory management as an MDP by defining states, actions, transitions, and rewards. Give the Bellman optimality equation for this MDP, dominant online complexity terms, and an evaluation protocol that separately measures retrieval, answer quality, citation faithfulness, calibration, latency, and data leakage.